Embedding Space Exploration

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* Introduction

The previous note explained what word embeddings are. This one explores them — taking real vectors and measuring what is actually inside the space.

All numbers below are measured from GloVe (glove-wiki-gigaword-100): 400,000 English words, each represented by 100 numbers.

1. Cosine Similarity Between Words

Cosine similarity measures the angle between two vectors:

$$\cos(A,B) = \frac{A \cdot B}{(|A| \times |B|)}$$

* Worked by hand:

Real vectors have 100 numbers, which is impossible to do on paper. With 2 numbers each the arithmetic is easy to follow, and it is exactly the same calculation.

Let $A = [2, 3]$ $B = [3, 1]$

$A \cdot B = (2 \times 3) + (3 \times 1) = 9$
$|A| = \sqrt{2^2 + 3^2} = 3.61$
$|B| = \sqrt{3^2 + 1^2} = 3.16$

$\cos(A, B) = \frac{9}{(3.61 \times 3.16)} = 0.79 \rightarrow$ similar

Now $C = [-3, 1]$

$A \cdot C = (2 \times -3) + (3 \times 1) = -3$
$|C| = \sqrt{(-3)^2 + (1^2)} = 3.16$

$\cos(A, C) = \frac{-3}{(3.61 \times 3.16)} = -0.26 \rightarrow$ unrelated

* Measured on real GloVe vectors

Four groups of words were chosen. If the embedding space is meaningful, words within a group should score high and words across groups should score low.

WITHIN A GROUP	COS	ACROSS GROUP	COS
cat - dog	0.88	cat - computer	0.30
computer - software	0.84	King - internet	0.25
King - prince	0.77	nepal - horse	0.09
King - queen	0.75	queen - software	0.04

The pattern holds cleanly: within-group scores are 0.75 - 0.88, across-group scores are 0.04 - 0.30. The space genuinely encodes meaning.

BUT LOOK CLOSER - ONE RESULT BREAKS THE PATTERN

nepal - india 0.68 both South Asian neighbors

nepal - france 0.22 both countries, yet low

Both pairs are "two countries", so a category-based system would score them the same. The embedding does not, because it learned from how the words are used, not from a category list. Nepal and India appear in the same news stories; Nepal and France rarely do.

Embeddings capture usage, not taxonomy.

2. Vector Arithmetic.

Relationships between words appear as directions, so they can be added and subtracted. To answer "man is to king as woman is to what?":

vector (King) - vector(man) + vector (woman) = ?

then find the word whose vector is nearest to the result

ARITHMETIC	NEAREST WORD	SCORE	RELATIONSHIP
King - man + woman	Queen	0.77	gender
paris - france + nepal	Kathmandu	0.81	country → capital
walking - walk + swim	swimming	0.80	verb tense
mice - mouse + cat	cats	0.77	singular → plural

four different kinds of relationship - gender, geography, grammatical tense and plurals - all recovered by the same simple arithmetic. Nobody programmed any of them in.

WHERE IT FAILS

ARITHMETIC	RESULTS
bigger - big + small	larger 0.89 · smaller 0.87 · large 0.79

The expected answer is smaller. It is found — but only in second place. The top answer, larger, keeps the comparative form but loses the reversal from big to small.

This is the same weakness seen in the previous note: the embedding places antonyms very close together, because opposites appear in the same contexts. Analogies work often, but not reliably.

A MORE SERIOUS PROBLEM — BIAS

doctor − man + woman   nurse 0.77 · physician 0.72 · doctors 0.68

The correct answer is doctor — the profession does not change with gender. The model returns nurse.

This is not a bug in the arithmetic. The vectors were trained on human-written text, and that text associates women with nursing and men with doctoring. The embedding faithfully learned a human bias. Any system built on these vectors — CV screening, search, translation — inherits it.

3. Visualizing the Space with PCA

Each word has 100 numbers, so the space cannot be drawn. PCA (Principal Component Analysis) squashes 100 dimensions down to 2 by keeping the directions along with the which the words differ most.

30 words were chosen from five clear categories — animals, royalty, countries, technology, and food — and projected to 2 dimensions.

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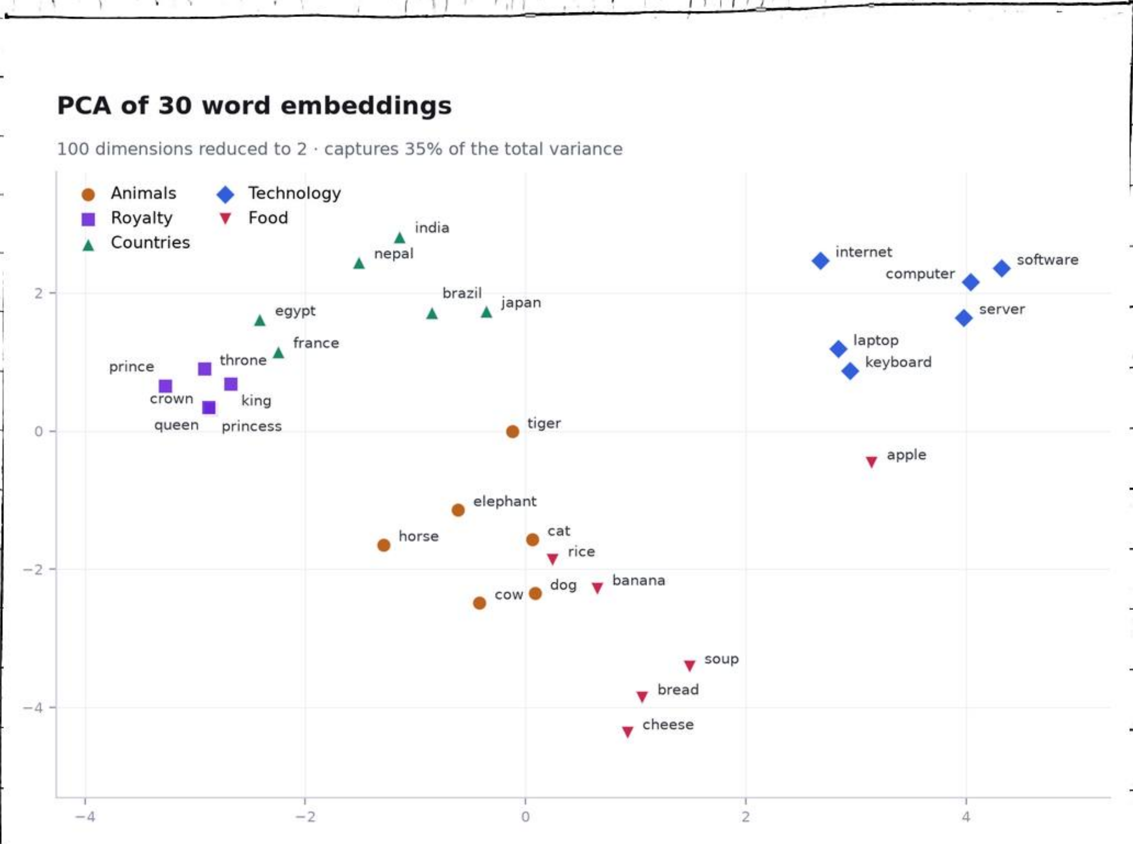

Figure 1: PCA
(Principal Component Analysis)

IMPORTANT CAVEAT

These 2 dimensions capture only 35% of the total variance. The other 65% of the information cannot be drawn Two words that look close on the page may be far apart in the real 100-dimensional space. The picture is a useful summary, never proof

4. Visualizing with t-SNE

t-SNE solves the same problem differently. PCA is linear — it finds the flattest sensible shadow of the data. t-SNE is non-linear and optimises for one thing only: keeping each word close to its true nearest neighbours.

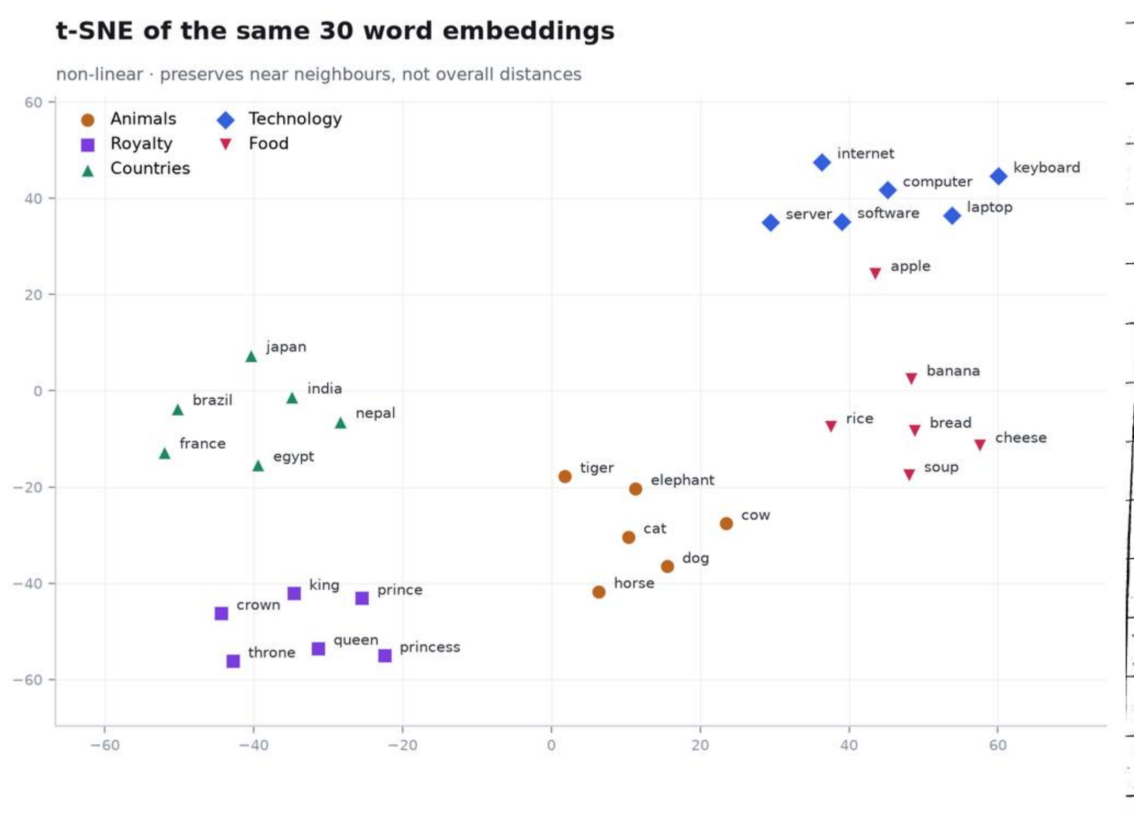

Figure 2: t-SNE

	PCA	T-SNE
Type	linear	non-linear
Preserves	overall variance	near neighbours
Distance between Clusters	meaningful	not meaningful
Same result every run	yes	no - depends on settings
Clusters look	Looser	tighter - better separated

t-SNE produces visibly cleaner groups. But its gaps must not be read as distances - a wide gap between two clusters does not mean they are especially unrelated.

5. Interpreting the Clusters

Both plots show the same five groups separating on their own. Nobody labelled these words. The grouping emerged purely from how the words are used in text - strong evidence that the embedding space encodes meaning.

Two finer patterns are visible inside the clusters:

- Countries sub-group by region. Nepal and India sit together, away from France and Brazil — matching the 0·68 versus 0·22 scores from 1
- Royalty is extremely tight. King, queen, prince, princess, thrown and crown almost overlap, because they appear in nearly identical contexts.

THE ANOMALY — ONE WORD IN THE WRONG PLACE

"apple" does not sit with the food cluster. In both plots it drifts towards computer, software and laptop.

The reason is that apple has two distinct meanings — the fruit and the technology company — and GloVe gives every word exactly one vector. That single vector has to serve both senses, so it lands in a compromise position between the two clusters, belonging properly to neither.

This is the fundamental limitation of static embeddings: one word, one vector, regardless of context. It is precisely the problem that contextual embeddings such as BERT were built to solve

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— they give apple a different vector in "apple pie" than in "Apple released a phone."

* Conclusion

The embedding space is genuinely meaningful: similarity scores separate related from unrelated words, arithmetic recovers gender & geography, tense, and plural relationships, and five categories cluster without supervision.

But the exploration also exposed three real limits. Analogies are unreliable (bigger - big + small ranks larger above smaller). The vectors carry human bias (doctor - man + woman returns nurse). And a single vector per word cannot represent two meanings, which is why apple sits stranded between food and technology.